Full-Stack Real-Time Voting Application Using Redux

Abstract

This project presents the development of a full-stack, real-time voting application built using Redux, React, and Node.js, following a test-first development methodology, where multiple users can participate in live voting sessions in which entries are compared in pairs, votes are cast, and results are displayed instantly without any manual page refresh. The system is architected as two separates but communicating applications — a React-based client serving both a mobile-friendly voting interface and a live results display, and a Node.js backend managing core voting logic — with real-time bidirectional communication handled through Socket.io over WebSocket's to ensure all connected users receive state updates simultaneously. Redux is employed for centralized state management on both the client and server sides, with Immutable.js maintaining predictable and consistent application state through pure functions organized as reducers and actions, while React components are connected to the Redux store to reflect live voting data reactively. The project further incorporates modern tooling including ES6, Babel, and Webpack, alongside Mocha-based unit testing to ensure code reliability, with the primary objective of demonstrating how a unidirectional data flow architecture powered by Redux and real-time WebSocket communication can be used to build scalable, maintainable, and responsive full-stack web applications with live multi-user functionality.

Team no: 18

Nisam Rahman k – Roll no: 44

P.S. Nandhana – Roll no: 45